Rank 1

Cluster 28 Cluster 51 Cluster 24 Cluster 93 Cluster 30 Cluster 36 Cluster 62 Cluster 48 Cluster 47 Cluster 32 Cluster 44 Cluster 16 Cluster 40 Cluster 41 Cluster 73 Cluster 66 Cluster 89 Cluster 128 Cluster 10 Cluster 49 Cluster 26 Cluster 67 Cluster 136 Cluster 18 Cluster 104 Cluster 69 Cluster 20 Cluster 57 Cluster 27 Cluster 46 Cluster 54 Cluster 100 Cluster 21 Cluster 98 Cluster 14 Cluster 86 Cluster 65 Cluster 82 Cluster 37 Cluster 81 Cluster 84 Cluster 137 Cluster 58 Cluster 97 Cluster 35 Cluster 129 Cluster 134 Cluster 39 Cluster 63 Cluster 109 Cluster 2 Cluster 52 Cluster 118 Cluster 71 Cluster 80 Cluster 50 Cluster 0 Cluster 61 Cluster 96 Cluster 102 Cluster 72 Cluster 131 Cluster 110 Cluster 53 Cluster 76 Cluster 42 Cluster 75 Cluster 45 Cluster 99 Cluster 95 Cluster 56 Cluster 113 Cluster 3 Cluster 55 Cluster 12 Cluster 79 Cluster 85 Cluster 126 Cluster 108 Cluster 29 Cluster 111 Cluster 133 Cluster 130 Cluster 101 Cluster 92 Cluster 94 Cluster 38 Cluster 6 Cluster 59 Cluster 74 Cluster 120 Cluster 116 Cluster 121 Cluster 125 Cluster 83 Cluster 43 Cluster 90 Cluster 88 Cluster 87 Cluster 124 Cluster 127 Cluster 117 Cluster 112 Cluster 11 Cluster 13 Cluster 9 Cluster 34 Cluster 106 Cluster 70 Cluster 68 Cluster 122 Cluster 132 Cluster 78 Cluster 114 Cluster 64 Cluster 33 Cluster 123 Cluster 8 Cluster 5 Cluster 119 Cluster 77 Cluster 31 Cluster 4 Cluster 17 Cluster 7 Cluster 19 Cluster 91 Cluster 115 Cluster 60 Cluster 1 Cluster 103 Cluster 22 Cluster 15 Cluster 135 Cluster 105 Cluster 107 Cluster 23 Cluster 25

Rank 3

Cluster 0 Cluster 1

Rank 4

Cluster 0 Cluster 1

Rank 2

Cluster 8 Cluster 10 Cluster 18 Cluster 20 Cluster 5 Cluster 6 Cluster 7 Cluster 16 Cluster 17 Cluster 21 Cluster 11 Cluster 2 Cluster 12 Cluster 13 Cluster 14 Cluster 0 Cluster 4 Cluster 3 Cluster 19 Cluster 1 Cluster 9 Cluster 15

Rank 5

Cluster 1

SETDB1
(1, 247)

CCND3
(2, 175)

ATR
(3)

KIF11
(3)

RB1
(3)

STAG1
(3)

BRD7
(4)

CDKN2A
(4)

PLK4
(5)

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(6, 23, 29, 30, 51, 65, 165, 181, 193)

CDKN2A
(7, 13, 131, 173, 198, 239, 282, 289)

GNAI2
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RAD21
(8)

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(9, 15, 102, 104, 127, 146, 150)

TP53
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KIF2A
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FANCD2
(102, 146)

PLK4
(127)

RAD21
(10, 109, 115, 196, 200, 212)

FANCD2
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GNAS
(11)

RAD21
(11)

STAG1
(11)

ATM
(12)

RAD21
(12)

SMAD4
(14, 128)

KIF2A
(16, 17, 26, 27, 37, 46, 55, 95, 96, 118, 163, 203, 236, 240)

STAG1
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SMC3
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KIF2A
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STAG1
(32, 38, 40, 41, 172)

GNAS
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STAG1
(44)

GNAS
(47)

NF1
(48, 293)

ATM
(49, 258)

KRAS
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STAG1
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ATR
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STAG1
(53, 211)

TP53
(53, 211)

ATR
(54)

NRAS
(54)

STAG1
(54)

CDKN2C
(56)

GNAS
(56)

TP53
(56)

CDH1
(58)

TP53
(58)

STAG1
(59)

TBL1XR1
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ATR
(60, 64, 138, 158, 184, 216, 253)

STAG1
(60, 64, 138, 158, 184, 216, 253)

CTNNB1
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PIK3CA
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ERBB4
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NRAS
(63)

KIF2A
(66, 74)

TP53
(66, 74)

KRAS
(67, 162, 189, 254, 255)

PIK3CA
(68, 160, 250, 299)

ARID1A
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TP53
(69, 290)

KEAP1
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STAG1
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TP53
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KIF2A
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SMAD4
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PRKAR1A
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ARID1A
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BRD7
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XPO1
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ATR
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KIF2A
(79)

USP44
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BRCA1
(81)

NF1
(81)

STAG1
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TGFBR2
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KMT2D
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ELF3
(87)

MYC
(89)

RAD21
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SRSF2
(89)

TP53
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KIF2A
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MEN1
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STAG1
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TERT
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SMC3
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STAG1
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NF2
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RB1
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FOXA1
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TP53
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BRAF
(99, 121)

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STAG1
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ERBB4
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VHL
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ARHGAP35
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CTNNB1
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WT1
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KLHL6
(114)

PLK1
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SMAD4
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STAG1
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PBRM1
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BUB1
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HLA-A
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PLK1
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ARID1A
(124, 134)

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FANCD2
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TP53
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RNF43
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SMAD4
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FANCD2
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KIF2A
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MEN1
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MGA
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PLK1
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TP53
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CDKN2A
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RAD50
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ATR
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CTNNB1
(141, 273)

STAG1
(141, 273)

SETD2
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EBF1
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BRCA2
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ARID1A
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KRAS
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FANCD2
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STAG1
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ARID1A
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RB1
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STAG1
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CDKN1B
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CTNNB1
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ATR
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STAG1
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FANCD2
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(156, 302)

NCOR1
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ERBB4
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MYD88
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RRAGC
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STAG1
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MAP2K4
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EBF1
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STAG1
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SETDB1
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TP53
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STAG1
(170, 243)

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TP53
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CCND3
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GNA13
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STAG1
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KMT2D
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BUB1
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PLK4
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TERT
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FANCD2
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ASXL1
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ATR
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SF3B1
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KIF2A
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ARID1A
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FOXA1
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